

Cell-averaged coupling for a cubic-voxel lattice: why the midpoint rule cannot be repaired by summing further

Abstract

A voxel lattice couples its cells through a propagator that is almost always *sampled at cell centres*. The cell has volume, so this is a midpoint rule, and its error is scale-invariant: refining the mesh shrinks the cell and the spacing together, so the bias never dilutes. This note establishes that the obvious repair — correcting neighbours one at a time out to some radius — *cannot be made to work*, because the correction has an $\mathcal{O}(1/R)$ tail and becomes summation-shape dependent beyond $kr \sim 1$, and the two requirements are incompatible at any box size. It then gives a closed-form replacement in two halves, exploiting the fact that the cube average is an identity on plane waves but not on a spatial sum. Against an exact reference the corrected scheme converges at second order where the truncated one saturates. A secondary consequence is the closure of a long-standing empirical constant: a fitted shear correction that had stood at 0.886234, then 1.023737, is measured here at 1.0000 and is shown to have been a propagator artefact throughout.

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1 The defect

Write the coupling between two cells as the propagator G evaluated at the separation of their centres. The source cell carries a volume moment, so the field its neighbour should receive is

$$\psi_i = \left[\frac{1}{V} \int_{V_{\text{source}}} G(\mathbf{x}_i - \mathbf{x}') d^3x' \right] \times (\text{moment}), \quad (1)$$

the average over the source cell, evaluated at the *receiver centre*. The usual practice replaces the bracket by $G(\mathbf{x}_i - \mathbf{x}_j)$: a midpoint rule.

The error is

$$\langle G \rangle - G = \frac{d^2}{24} \nabla^2 G + \mathcal{O}(d^4), \quad (2)$$

with d the cell side. What makes this pernicious rather than merely inaccurate is that it is *scale-invariant on a space-filling lattice*. The cells and their separations shrink together, so the relative error at each neighbour is unchanged by refinement, and the summed bias tends to a constant that no mesh refinement removes.

Which average. Equation (1) averages over the source cell only, with the receiver a point. That is the choice consistent with a collocation scheme, whose state variable is a point value at the cell centre. A Galerkin scheme, whose state is a cell average, wants a *double* average over both cells. The two differ, and the distinction is load-bearing: on this solver the collocation pairing outperforms the Galerkin one by factors of 3.4 and 16 against an exact reference. For an ellipsoid the two coincide, by Eshelby’s uniformity theorem; a cube is the only place they can differ at all.

2 Why the obvious repair fails

The natural fix is to correct neighbours directly out to some radius R and accept the midpoint rule beyond it. This fails for two reasons that are individually survivable and jointly fatal.

2.1 The tail is long-ranged

Away from the origin the propagator entries satisfy the Helmholtz equation, $\nabla^2 g_c = -k_c^2 g_c$ exactly, and derivatives commute with ∇^2 . Two regimes follow from (2):

- $kr \ll 1$: the elastostatic Kelvin tensor is *biharmonic*, not harmonic ($\nabla^2 r = 2/r$), so $\nabla^2 G \neq 0$ but sits two orders below G . With the displacement block $G \sim 1/r$ this gives a correction $\sim d^2/r^3$, a shell sum $\sim d^2/R^2$, and an $\mathcal{O}(1/R)$ tail — slow, but absolutely convergent.
- $kr \gg 1$: Helmholtz makes the correction $\rightarrow -(kd)^2 G/24$, so it *tracks the kernel itself*. Each shell then contributes a constant and only oscillation converges the sum — conditionally.

Both halves are measured. The far-field ratio $|\langle G \rangle - G|/|G|$ plateaus at 9.14×10^{-4} against the predicted $(k_S d)^2/24 = 9.375 \times 10^{-4}$, within 3%; and the static ratio falls as $1/r^2$ exactly, the signature of a surviving $d^2 \nabla^2$ term rather than a cancelled one.

2.2 The far field is shape-dependent

Summing the correction over a *square* region and over a *circular* one tests conditional convergence directly. At $k_S r_{\max} = 0.4$ the gap between them shrinks by 5.4×: absolutely convergent. At $k_S r_{\max} = 3$ it *grows* by 2×: conditionally convergent, so the limit depends on the shape of the region summed.

2.3 The two are incompatible

Converging the $\mathcal{O}(1/R)$ tail requires $R \sim 80$ cells. The shape term switches on near $R \sim 1/(k_S d) \approx 50$ cells. *The box cannot be made large enough to converge without becoming large enough to be shape-dependent.*

No radius is correct. This is a structural obstruction, not a tuning problem, and it is why the truncated correction had to be replaced rather than extended.

3 The closed-form replacement

The cell average is a convolution with the cell indicator, so on a plane wave it is exactly a multiplication by the cube form factor

$$f_0(\mathbf{k}) = \text{sinc}(k_1 h) \text{sinc}(k_2 h) \text{sinc}(k_3 h), \quad h = d/2, \quad (3)$$

one power — the single average of (1). The squared form factor $|f_0|^2$, which appears in the existing literature and in this package’s own inter-voxel tables, is the *double* (Galerkin) average: source cell convolved with field cell.

A form factor multiplies plane waves, and nothing else. This is the organising fact, and it splits the construction in two. Where the lattice kernel is a plane-wave sum, the average is an identity. Where it is a spatial summation, no form factor exists.

3.1 Between planes: an identity

For non-equal depths the Bloch kernel is a plane-wave sum, and (3) applies directly. Two properties of the implementation are worth recording because neither is optional.

Per mode. The P and S poles carry different k_3 , so a single scalar factor on the assembled block merely rescales it; each pole must be scaled by the sinc built from its own k_3 .

The decay margin halves. On the evanescent branch $k_3 = i\kappa$ and

$$\text{sinc}(i\kappa h) = \frac{\sinh \kappa h}{\kappa h} \sim \frac{e^{\kappa h}}{2\kappa h}, \quad (4)$$

which *grows*. Against the kernel’s own $e^{-\kappa|\Delta z|}$ the product decays as $e^{-\kappa(|\Delta z|-h)}$: still convergent, because $|\Delta z| \geq 2h$ for distinct planes, but at half the rate at adjacent planes. A reciprocal-sum cutoff tuned for the point kernel silently returns 10^{-8} where it had returned 10^{-16} .

3.2 In-plane: an analytic tail

At equal depth the route is an Ewald sum, whose real-space half is a spatial summation — so no form factor applies. Nor can the plain reciprocal sum be substituted: with the *single* form factor its terms decay as $1/|\mathbf{G}|^2$ against $\sim |\mathbf{G}|$ growth of the strain block and $\sim |\mathbf{G}|$ states per shell, which is logarithmically divergent. (The double form factor gains a further $1/|\mathbf{G}|^2$ and does converge, which is why the squared route never met this obstruction.)

Instead, using $\nabla^2 g_c = -k_c^2 g_c$ exactly away from the origin,

$$\sum_{|\mathbf{R}| > R_0} \langle D \rangle = \left(1 - \frac{k_c^2 d^2}{24}\right) (D_{\text{full}} - D_{\text{near}}) + \mathcal{O}(d^4), \quad (5)$$

where D_{full} is the existing Ewald sum — already computed once per mode, which is exactly the granularity (5) needs, since P and S carry different k^2 . The near shell is averaged directly; the far field is closed form.

The sharp test is R_0 -independence. R_0 splits the sum between the two pieces, so a wrong tail would make the answer drift with it. Measured drift is 3.2×10^{-6} and 3.8×10^{-6} for $R_0 = 3, 4$ against $R_0 = 2$.

case	scheme	error ratio per mesh halving	
		$n_z:2 \rightarrow 4$	$4 \rightarrow 8$
pure shear	truncated shell	2.2	1.29
	closed form	6.1	3.70
mixed	truncated shell	2.0	1.19

Table 1: The truncated scheme *saturates* — its ratio tends to 1.0, a floor — because its bias is scale-invariant. The closed-form scheme tends to 4.0, second order.

4 Results

Against Kennett, exact for a uniform layer and independent of the construction:

The single-mesh error is the least informative number here; the distinction is between a scheme that keeps improving under refinement and one that stops. The mixed case’s closed-form column is omitted deliberately: its error passes through zero between $n_z = 4$ and $n_z = 8$, so no convergence order can be quoted for it. Its magnitude is below the truncated scheme’s at every mesh tested.

Validity. The in-plane tail is an expansion in $(\kappa d)^2$. Measured across frequency it is flat to $k_S d \approx 1$ and degrades beyond; a larger near shell reduces the error at every frequency tested, confirming that the expansion and not something else is the limiting term. The regime of use, $ka < 0.3$, sits an order of magnitude inside that.

5 A fitted constant, closed

A one-parameter shear correction K had been carried for some time, defined so that $K = 1$ means the cube’s own shear response is correct. Its history:

value	context	what it was measuring
0.886234	original	a doubly-averaged contact term against a collocation state
1.023737	contact fixed	the remaining average truncated at a finite shell
1.0000	this work	nothing

Measured on the corrected propagator, $|K - 1| = 2.75 \times 10^{-4}$. Each time a propagator defect was removed, K moved toward unity; there is no mechanism to derive. The near-misses $8/9 = 0.888889$ and $\sqrt{\pi}/2 = 0.886227$ were close to an artefact.

The methodological point is the durable one. A one-parameter correction fitted against the same reference it is then scored on will look successful: K was universal across contrasts, did nothing to the channels it should not have touched, and improved cases it was never fitted to. It was still an artefact. Relatedly, the tempting success metric — how much the answer improves once the parameter is tuned — reaches $290\times$ here and means nothing, because one knob with leverage cancels almost any small scalar residual. The honest measure is the distance of the *unfitted* prediction from ideal.

6 What is not established

1. **The in-plane half is still an expansion.** Equation (5) is exact through d^2 ; the between-plane half is exact at any cell size. This is the one place a small- d limit re-enters a formu-

lation that otherwise avoids them, and it is the origin of the frequency degradation above. Section 7 sets out what an exact version would require.

2. **The dynamic verification is incomplete.** The spectral single average was verified against a real-space reference in the static limit, where the ultraviolet behaviour in question lives. The dynamic case, with its poles and the subtracted formulation they force, is untested.
3. **One convergence order is unpinned,** as noted above.
4. **The isolated cube is not the lattice.** Against a subdivided-cube reference that resolves the internal field, the uniform-strain single-site T-matrix differs by $\approx 0.35\%$, while its effective response in a space-filling lattice is correct to 2.75×10^{-4} . Both hold: the discrepancy largely cancels among identical neighbours. The 0.35% is *not* decisive — changing the sub-cell coupling moves it by four times its own size — and sharpening it requires a coupling whose bias is below 10^{-3} , not a richer single-site basis.

7 Making the in-plane average exact

The obstruction is stated precisely in (5): the Ewald real-space half is a spatial sum, so the form factor does not apply, and the plain reciprocal sum is logarithmically divergent under the single form factor. The d^2 expansion is a way around that, not a resolution of it.

Averaging is linear, so it distributes across the Ewald split:

$$\sum_{\mathbf{R}} \langle G \rangle(\mathbf{R}) e^{-i\mathbf{k} \cdot \mathbf{R}} = \langle \text{real-space screened half} \rangle + \langle \text{reciprocal half} \rangle. \quad (6)$$

Each half admits an exact treatment, by different means:

- **Reciprocal half.** Plane waves, so the form factor multiplies. The vertical factor is not sinc at equal depth, because the average crosses the $|z|$ branch point at the cell centre, but it remains in closed form:

$$\frac{1}{2h} \int_{-h}^h e^{i\gamma|u|} du = \frac{e^{i\gamma h} - 1}{i\gamma h}. \quad (7)$$

- **Real-space half.** The screened kernel is short-ranged and regular at every non-self lattice site — the nearest in-plane neighbour has $|\mathbf{r}| \geq d - h\sqrt{3} > 0$, and $\mathbf{R} = 0$ is excluded as the self term — so ordinary quadrature averages it exactly. No form factor is needed, and none exists.

This would remove the last small- d expansion from the construction and with it the frequency degradation, at the cost of reworking the Ewald internals rather than wrapping them. It is not implemented here, and the argument above is a derivation sketch, not a validated result.